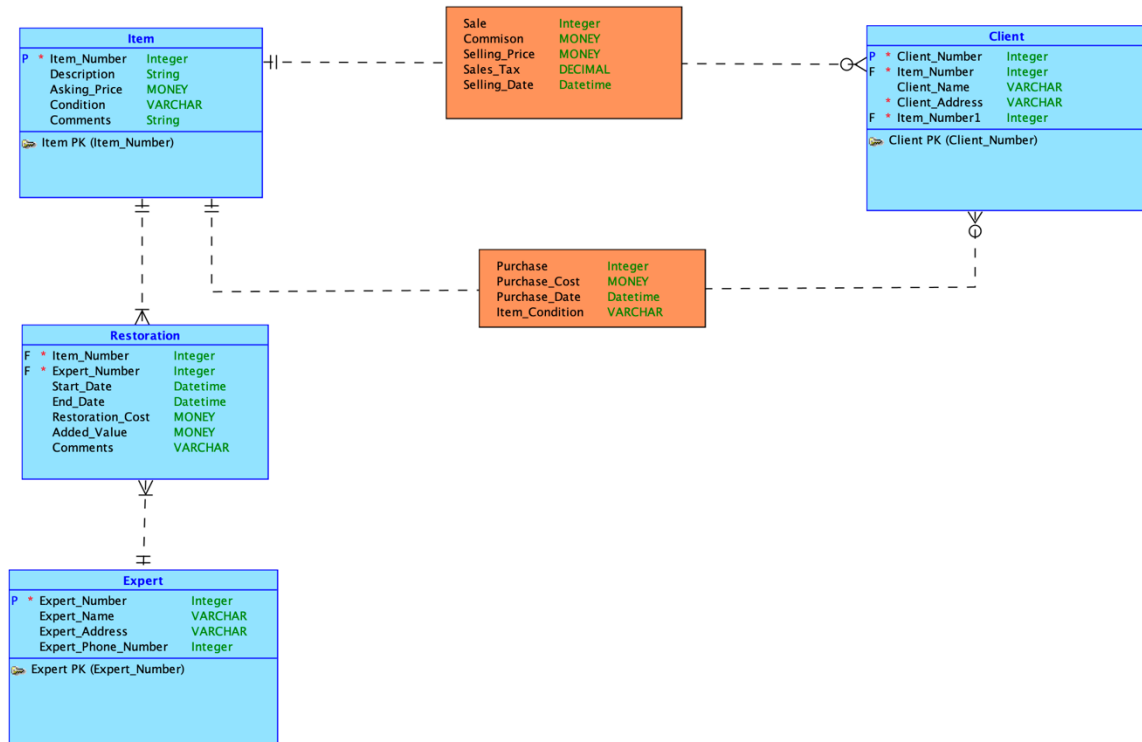




1. Your client received news of a new expert moving into town and they are considering contacting them for possibly restoring some of the items, so your client needs to store the contact info for this expert into the database. Does your ERD allow for this?

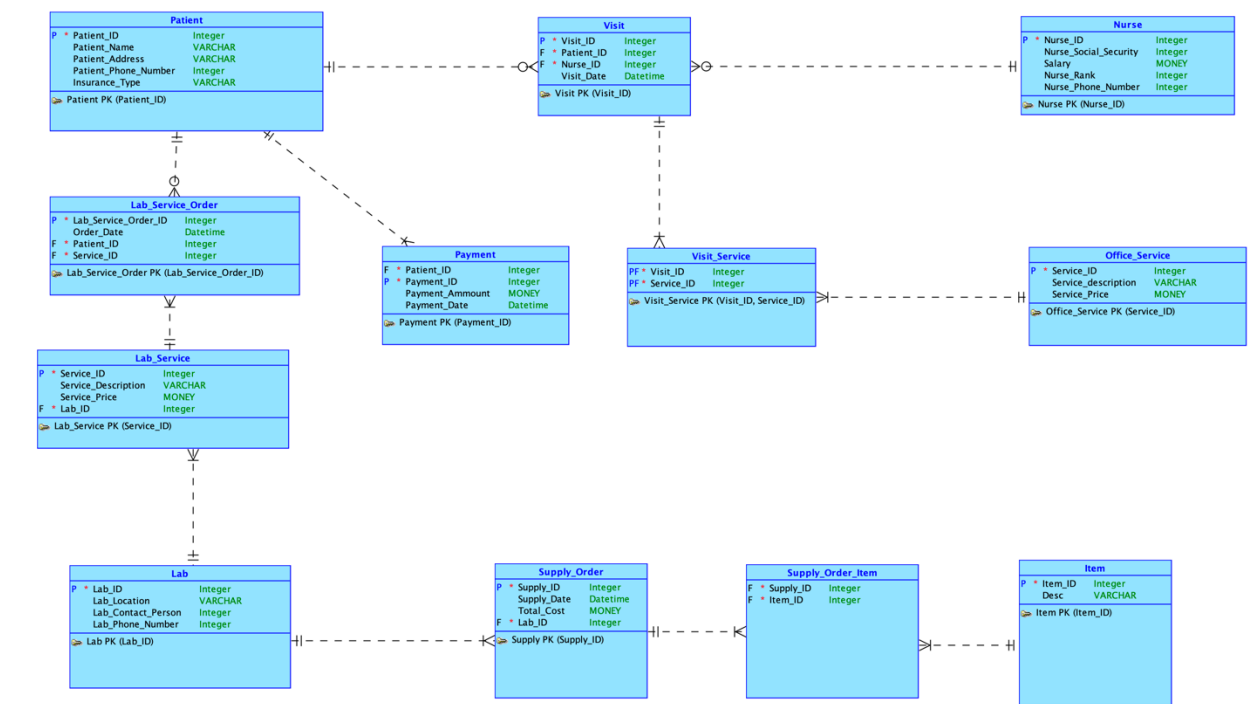
Yes, the expert is stored as its own entity with the primary key or Expert_Number this means that the expert they are thinking about adding can be, without being related to an item or a restoration

2. Some items belong in a set, although they can be sold separately. Does your ERD store this information? If not, how can you modify it, so this is possible?

It does not allow this; it can be modified to allow this by adding an attribute to item called set number and this would store what set it is a part of. This way it keeps track of if the item is a part of a set, but it still allows it to be sold on its own and not as a set.

3. Some items gain and lose value very quickly, depending on trends and availability. Your client wants to record the price history for some of these hot items, so that they can later analyze trends. Can you modify your ERD to accommodate this?

Yes, it can be modified by adding a price history entity, inside this entity you can track the item number, original price, price increases, price decreases and the dates of all of these. This would be a 1 to many relationship and would allow for the customer to track the prices of a specific item over time



1. How would you change your diagram if the cost of the service is not set? Frequently these services are personalized, and the cost varies by case, determined for example by the time required for the completion (i.e. the service has a price that is \$x per hour, and you also need the time required to calculate the cost for each patient)?

You would just need to clarify that the service Price is now a rate and not a flat fee and a visit service should track the start and end time of the service. The total cost of the service can then be derived from those values, and stored in the database and that could be stored in the visit service table

2. How would your diagram change if there are more than 1 dentists working in the office? You could then need to know who the dentist for each visit is and who is the dentist who is performing each of the services (not the lab services).

I would need to create a table that stored the values of all the dentists such as their name and phone number all the identifiers. The dentist table would then have to be linked to visit via 1 to many. There would also have to be a 1 to many relationship for the visit Services table since that can must also be kept track of.

3. Now nurses have different ranks, ad some nurses supervise others. Can you change your diagram to take this into account?

For each nurse a rank can be added to their table. Now that nurses would be able to supervise other nurses this can be done with a unary relationship where it points back to another nurse, it would also need a supervisor_ID attribute where it points to the nurse who is the supervisor,